جامعة ٦ أكتوبر
October 6 University

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Salivary Biomarkers In Cancer Detection

Non-Invasive Liquid Biopsy · Early Oral Cancer Screening · Salivary Diagnostics

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~3,000 Proteins
in human saliva

0 Needles
Zero Discomforts

95% Sensitivity
oral cancer detection

10 min Result
Biosensor Speed

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01 What are Salivary Biomarkers ?

Measurable molecular signals in saliva indicating cancer — before symptoms appear

DNA/cfDNA
- TP53 & KRAS mutations
- Methylation patterns
- Cell-free tumor DNA
- Detected by ddPCR in saliva

Proteins
- IL-8: 10x elevated in OSCC"
- SAT1 enzyme overexpressed
- HER2 antigen
- CYFRA 21-1
- Direct protein biomarkers

microRNA
- miR-31 (OSCC-specific)
- miR-200a, miR-184
- Stable encapsulated in exosomes
- Highly cancer-specific

Metabolites
- Polyamines altered in cancer
- Organic acid shifts
- Lipid metabolite profiles
- Warburg effect byproducts

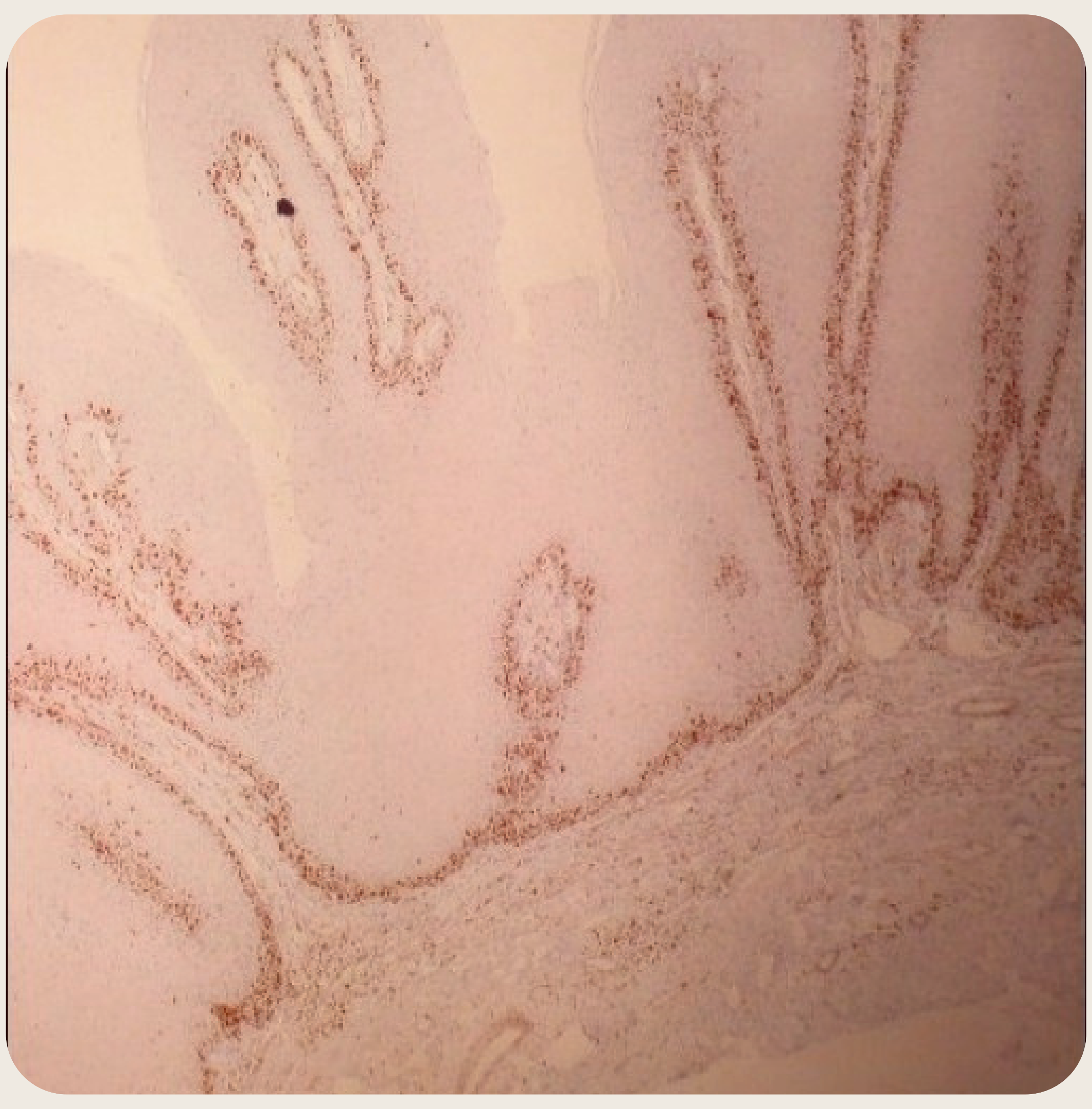


Figure 1. p53 IHC expression in oral leukoplakia. Basal and focal suprabasal nuclear staining. Mean LI = 75 ± 16.83%. Suprabasal p53 extension, visible here, is a critical early marker: Cruz et al. reported that 86% of premalignant lesions with suprabasal p53 expression developed into invasive SCC. This histological feature directly corresponds to the salivary cfDNA and mRNA elevations detected in high-risk oral dysplasia patients

How Signals Reach Saliva?

TUMOR
Sheds cfDNA
→
Blood
Carries it
→
Gland
Secrets
→
Saliva
Detachable

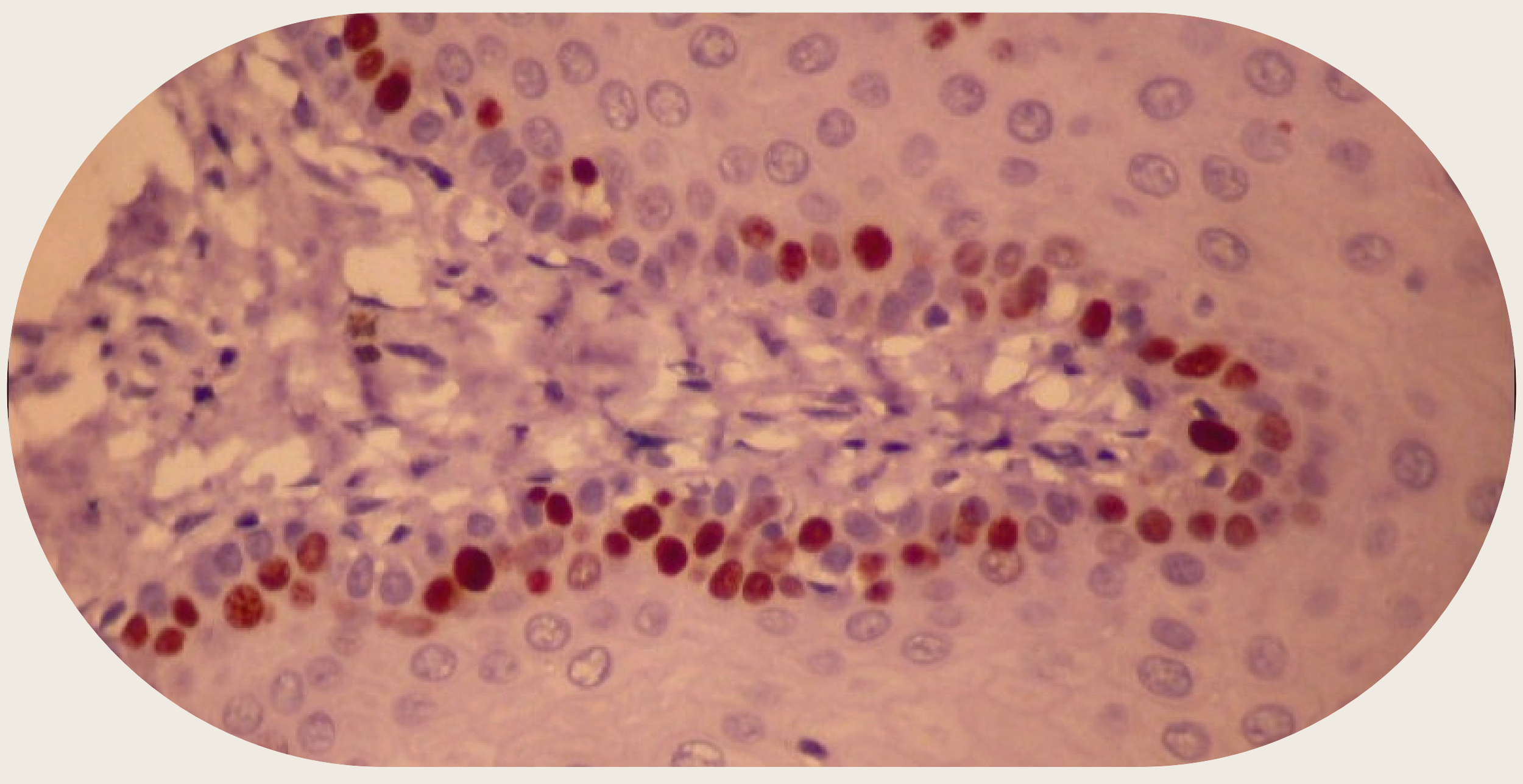


Figure 2. Ki-67 IHC expression in oral submucous fibrosis (MIB-1). Intense basal layer staining. Mean LI = 53.75 ± 39.66%. High inter-case variability (SD ±39.66) reflects heterogeneous proliferative activity, two OSMF cases showed LI >80%, indicating active epithelial hyperplasia with malignant potential, while two showed LI <25%. This variability explains the wide range of salivary mRNA levels reported in OSMF patients across different studies [11].

02 Detectable Cancers via Salivary Biomarkers

Oral SCC
Sensitivity —>95%
Specifity —-> 91%
IL-8 · miR-31 · SAT1
cfDNA · CYFRA 21-1
Direct shedding

Breast
Sensitivity —>76.7%
Specifity —-> 94.9%
KCNJ3 mRNA · HER2
miR-21 · miR-155
Via bloodstream

Lung
Sensitivity —> 88.5%
Specifity —-> 88.5%
Haptoglobin · AZGP1
Calprotectin · EGFR
Via bloodstream

Panceatic
Sensitivity —>90%
Specifity —-> 95%
KRAS · CA19-9
cfDNA methylation
Earliest detected

03 Detection Methods

Elisa
Protein Quantification
- Sandwich antibody assay
- Detects IL-8, HER2, CYFRA 21-1
- Femtomolar sensitivity
- Gold standard for proteins

Biosensor
Point-of-Care Device
- Graphene electrode chip
- Result in 8-15 minutes
- Smartphone-readable output
- Designed for the dental chair

qPCR / NGS
Nucleic Acid Analysis
- Quantifies mRNA & microRNA
- TP53, KRAS mutations in cfDNA
- 97.9% specificity for OSCC
- Single-molecule ddPCR precision

AI / ML
Multi-Omic Integration
- Combines all biomarker classes
- AUC > 0.95 for oral cancer
- 34% fewer false positives
- Cloud-based analysis platform

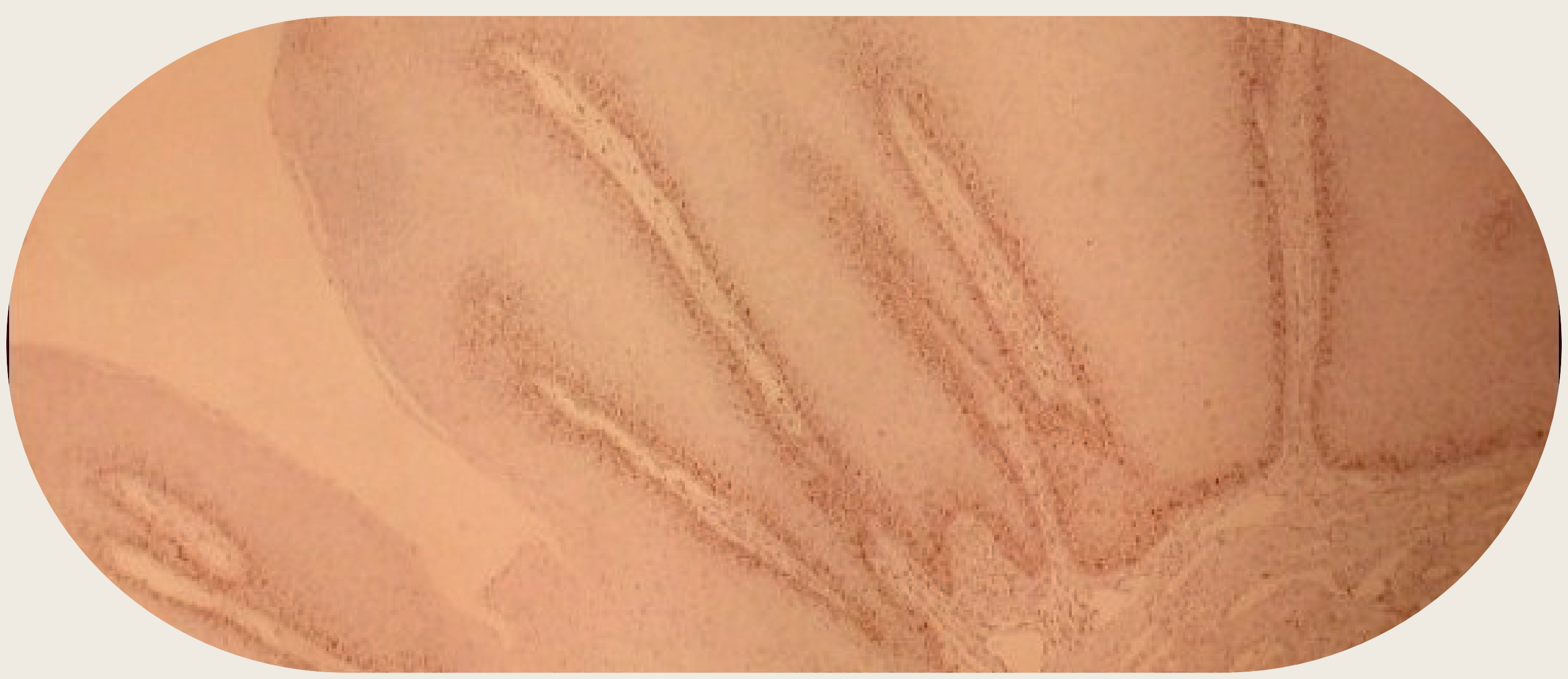
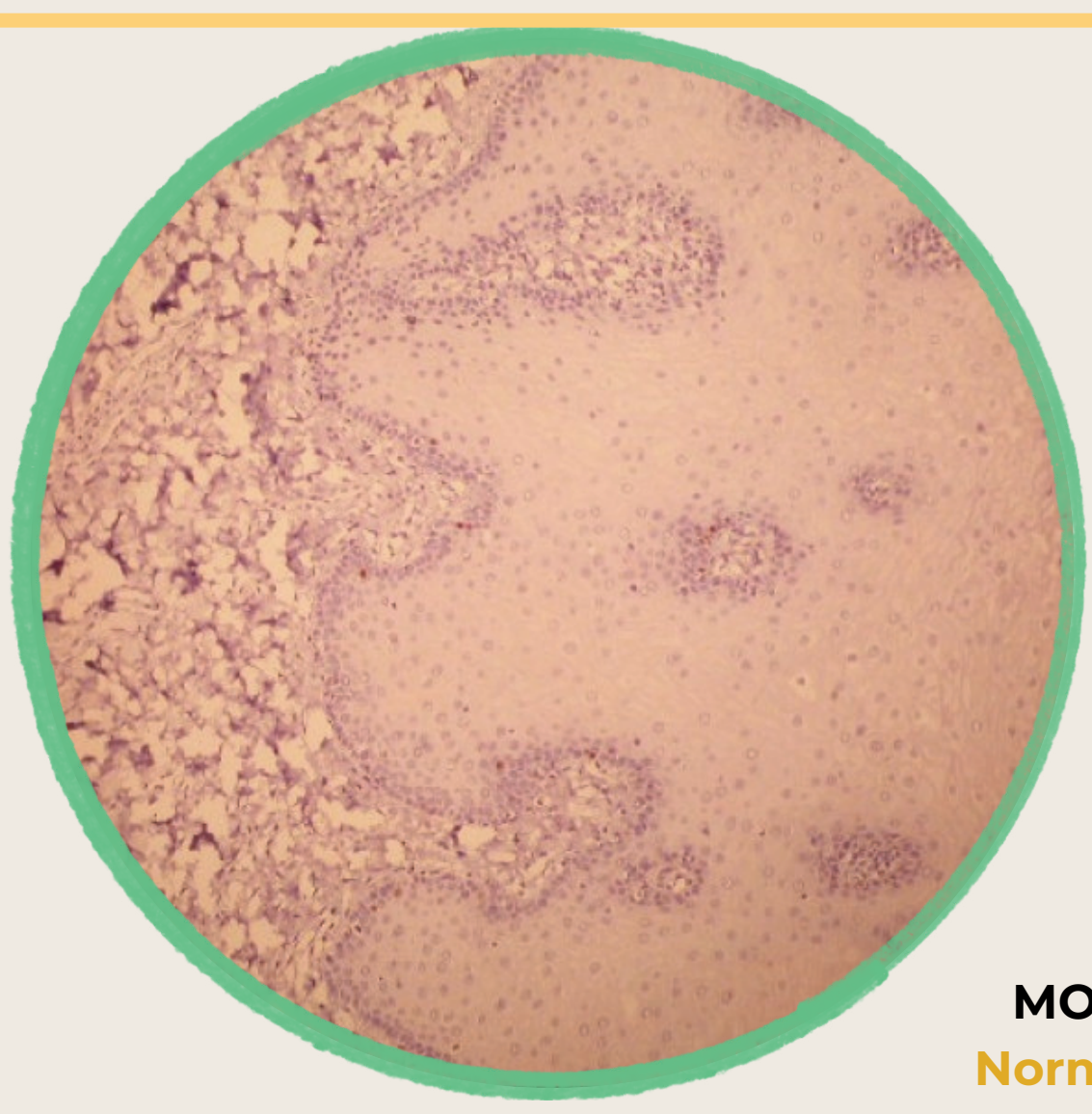


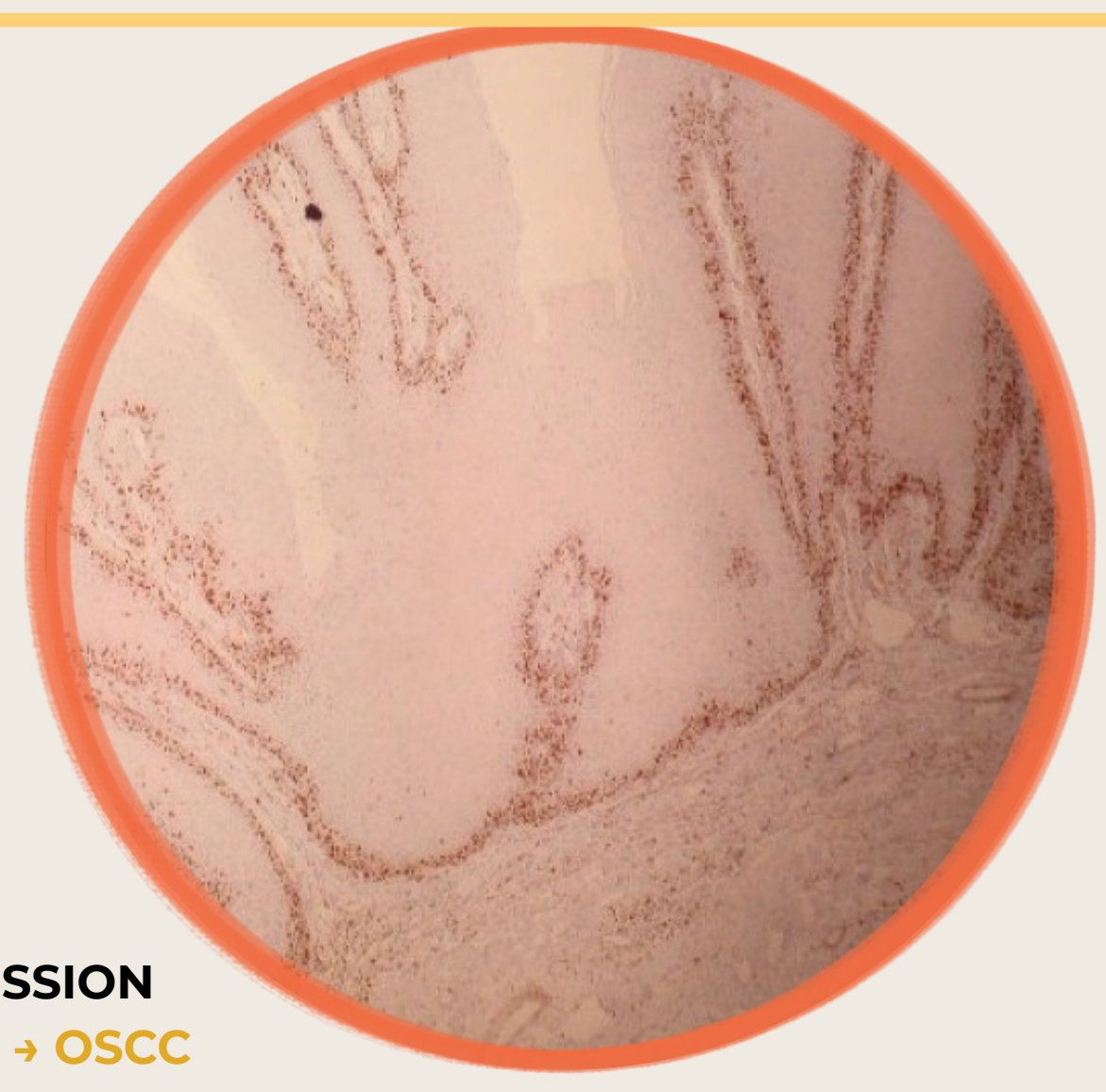
Figure 10. Ki-67 IHC expression in OSCC (MIB-1). Pan-epithelial staining extending through all layers. Mean LI = 25 ± 20.82% in well-differentiated OSCC. Importantly, Ki-67 LI increases with dedifferentiation: moderately differentiated OSCC shows LI 47 ± 6.3%; poorly differentiated OSCC shows LI 60 ± 2.8% [11]. This Ki-67 grading correlation directly predicts the magnitude of the salivary biomarker signal, poorly differentiated tumors generate stronger liquid biopsy signals.

Immunohistochemical Evidence — p53 & Ki-67 Baseline vs Disease



p53 · Normal Oral Mucosa
LI = 20% · Basal layer only
Physiological stress response

MOLECULAR PROGRESSION
Normal → Premalignant → OSCC
Each stage detectable in saliva by cfDNA & mRNA



Ki-67 · OSCC — All Layers
LI = 25% well-diff → 60% poorly-diff
Pan-epithelial: marks transformation

04 Advantages vs Limitation

Advantages	limitations
Zero needles	Lower concentrations
Real Time Monitoring	Contamination risk
Early detection	Can Not replace biopsy
Chariside deployments	Standrization

05 Systematic reviews in Egypt and worldwide

Owecki et al. 2025
LDH
MMP-9
Chemerin
33 articles

LDH SMD. 4.6
MMP-9 SMD. 1.51
Chemerin SMD 3.9

Large effect sizes · all p < 0.001

Wijaya et al. 2026
Mixed proteins
18 studies
1,647 pts

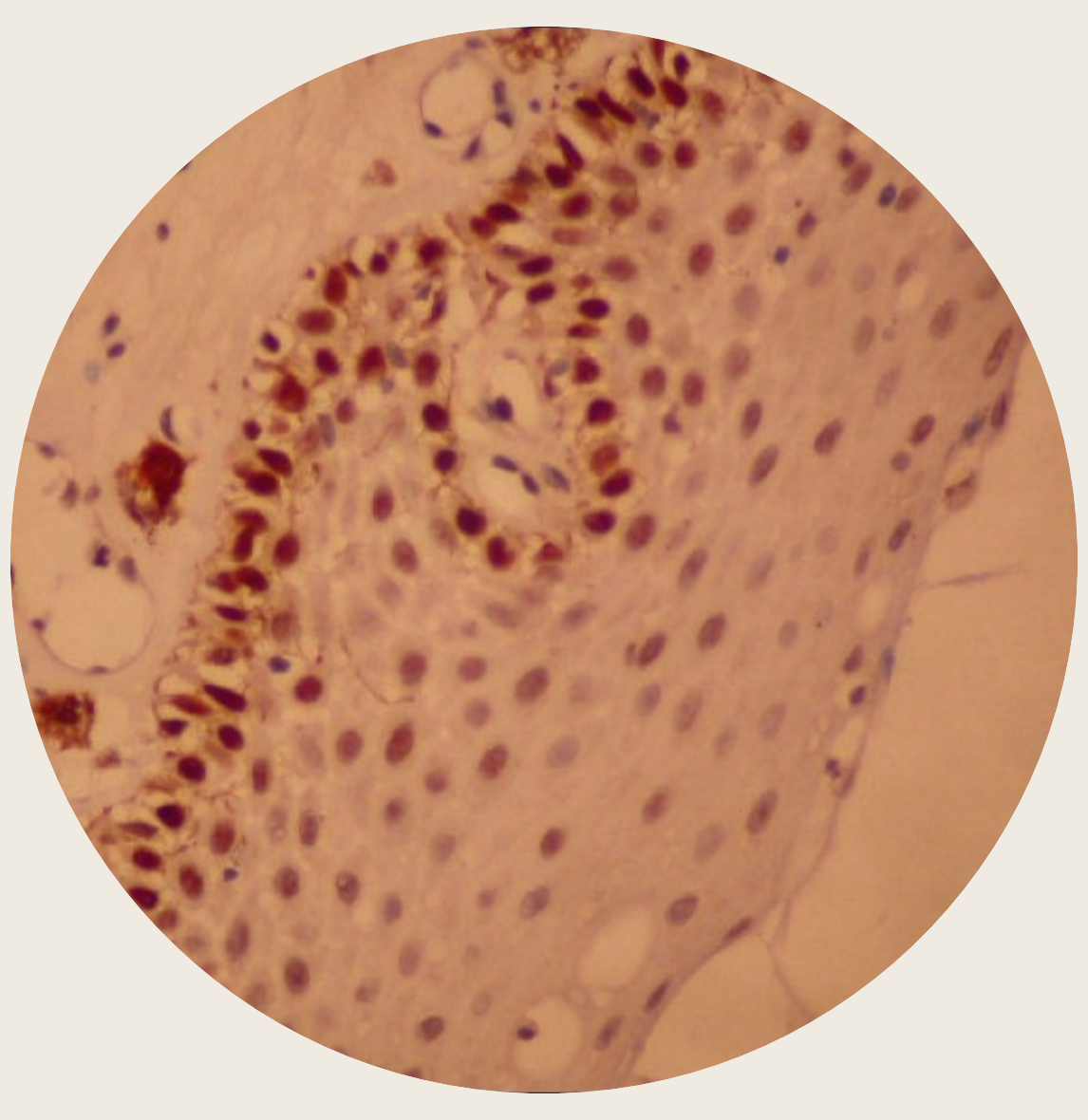
Sensitivity 0.64
Specificity. 0.71
AUC. 0.75

High heterogeneity · adjunctive triage use

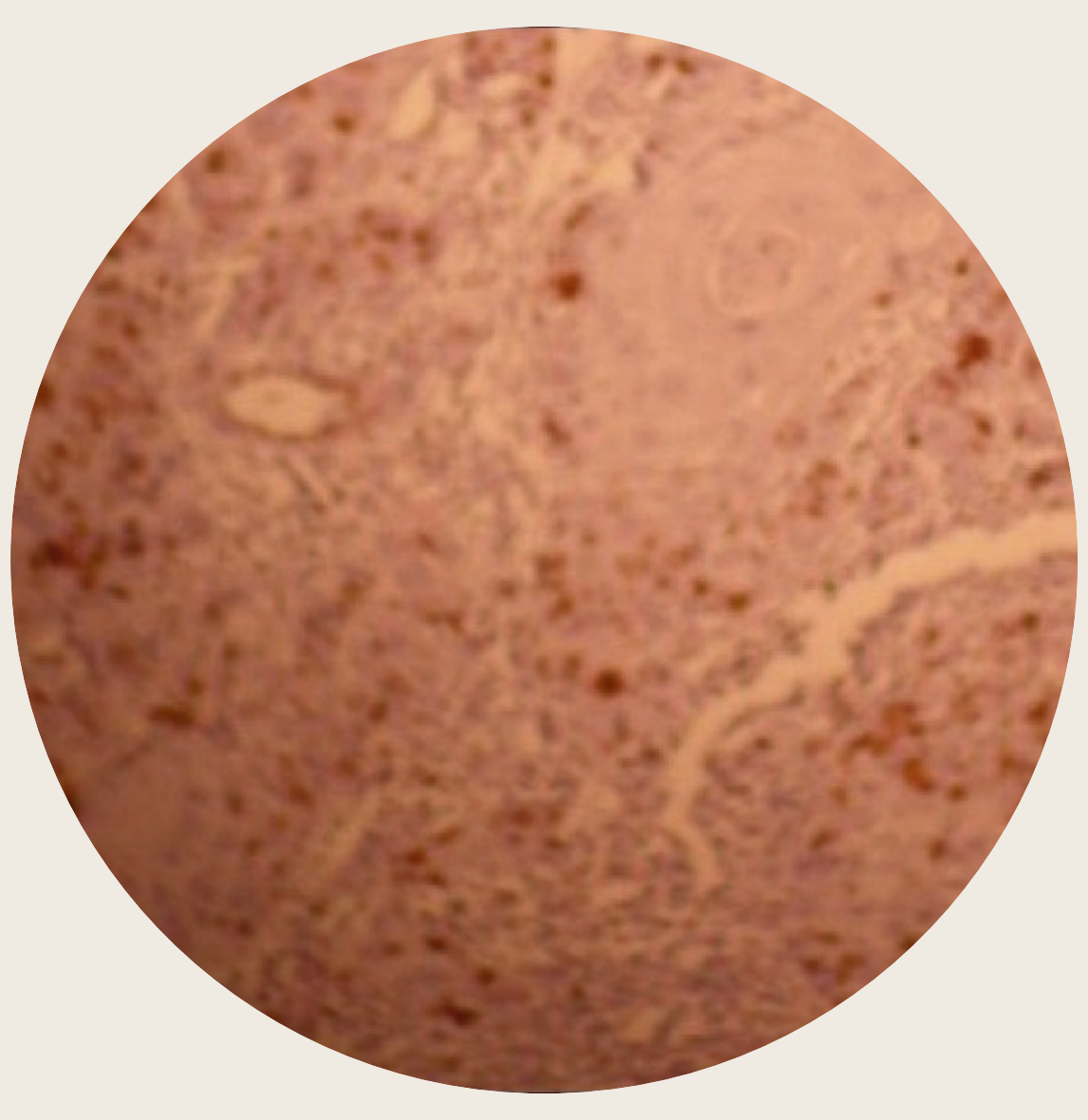
Shaukat et al. 2026
Salivary MMP-9
7 studies
795 pts

Sensitivity 0.98
Specificity. 0.96
AUC. 0.97

Highest single-marker accuracy reported



p53: Strong basal p53 overexpression signals early genetic damage before histological dysplasia, correlating with salivary cfDNA.



Ki-67: Suprabasal Ki-67 expansion indicates abnormal proliferation linked to early salivary biomarker elevation.

Conclusion

Saliva is a liquid biopsy, encoding cancer's molecular signature before symptoms appear. p53 and Ki-67 IHC confirm molecular changes begin at the premalignant stage, detectable in saliva long before clinical diagnosis, without a single needle.

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